Topics in Error-Correcting Codes

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Lecture 11: Polynomial Codes

For this lecture, we broadly follow Prahladh Harsha's notes, linked below [1].

Reed–Muller Codes

Fix a finite field $\mathbb{F}_q$, a number of variables $m \geq 1$, and a degree bound $d \geq 0$. The *total degree* of a monomial $X_1^{e_1} \cdots X_m^{e_m}$ is $e_1 + \cdots + e_m$; the total degree of a nonzero polynomial is the largest degree of a monomial with nonzero coefficient.

On $\mathbb{F}_q$, the identity $X^q = X$ allows us to reduce each individual degree to at most $q - 1$ without changing any evaluations or increasing total degree. We therefore work with *reduced* polynomials

$$P(X_1, \ldots, X_m) = \sum_{\substack{0 \leq e_i \leq q-1 \\ e_1 + \cdots + e_m \leq d}} c_{e_1,\ldots,e_m} X_1^{e_1} \cdots X_m^{e_m}.$$

Definition 1 (Reed–Muller code)**.** The code $\mathrm{RM}_q(m, d)$ consists of the evaluation vectors

$$\mathrm{Enc}(P) = \big(P(a)\big)_{a \in \mathbb{F}_q^m},$$

where $P \in \mathbb{F}_q[X_1, \ldots, X_m]$ is reduced and has total degree at most d. We fix an ordering of $\mathbb{F}_q^m$.

Proposition 2 (Length and dimension)**.** *The code* $\mathrm{RM}_q(m, d)$ *is* $\mathbb{F}_q$*-linear, has length* $n = q^m$*, and has dimension*

$$K_q(m, d) = \#\left\{ (e_1, \ldots, e_m) \in \{0, \ldots, q-1\}^m : \sum_{i=1}^m e_i \leq d \right\}.$$

Proof. Evaluation is linear. To show that the allowed monomial evaluations are independent, it suffices to prove that a reduced polynomial vanishing on all of $\mathbb{F}_q^m$ is zero. For $m = 1$, this is the univariate root bound. In general, write the polynomial as a polynomial in X_m with coefficients in the other variables. After fixing those variables, the resulting polynomial has degree at most $q - 1$ and vanishes at all q field elements, so every coefficient vanishes. Induction on m now shows that each coefficient polynomial is zero. Hence evaluation is injective, and the dimension is the number of allowed monomials. □

In particular, the q^m reduced monomials form a basis for all functions $\mathbb{F}_q^m \to \mathbb{F}_q$. Thus every such function has a unique reduced polynomial representation. The interesting degree range is $0 \leq d \leq m(q - 1)$.

The Low-Degree Case

Assume $0 \le d < q$. The individual-degree constraints impose no additional restriction, so stars and bars gives

$$K_q(m,d) = \binom{m+d}{d}.$$

For $m = 1$ we recover the Reed–Solomon code of degree at most d evaluated at all q field elements. For $m = 2$ and $d = 2 < q$, a message specifies the six coefficients of

$$P(X,Y) = c_{00} + c_{10}X + c_{01}Y + c_{20}X^2 + c_{11}XY + c_{02}Y^2.$$

Its codeword contains q^2 evaluations. Notice that a total-degree bound of 2 excludes X^2Y^2, even though each individual degree is 2.

The Schwartz–Zippel lemma

The univariate root bound says that a nonzero degree-d polynomial has at most d roots. In several variables, a polynomial can vanish on a whole hyperplane. The right statement bounds the *fraction* of zeros.

Lemma 3 (Schwartz–Zippel / polynomial distance lemma)**.** *Let $P \in \mathbb{F}[X_1, \dots, X_m]$ be a nonzero polynomial of total degree at most d, and let $S \subseteq \mathbb{F}$ be finite and nonempty. Then*

$$\Pr_{a \in S^m}[P(a) = 0] \le \frac{d}{|S|}.$$

Proof. If $d \ge |S|$, the bound is immediate, so assume $d < |S|$. Induct on m. For $m = 1$, this is the univariate root bound. Write

$$P(X_1, \dots, X_m) = \sum_{j=0}^{r} P_j(X_1, \dots, X_{m-1}) X_m^j, \qquad P_r \neq 0.$$

Since $\deg P_r \le d - r$, the induction hypothesis gives

$$\Pr_{a' \in S^{m-1}}[P_r(a') = 0] \le \frac{d-r}{|S|}.$$

If $P_r(a') \neq 0$, then $P(a', X_m)$ is a nonzero univariate polynomial of degree r, so it vanishes at at most r choices of $a_m \in S$. Splitting according to whether $P_r(a') = 0$ gives

$$\Pr[P(a) = 0] \le \frac{d-r}{|S|} + \frac{r}{|S|} = \frac{d}{|S|}.$$

□

Corollary 4 (Low-degree Reed–Muller parameters)**.** *For $0 \le d < q$, the code $\mathrm{RM}_q(m,d)$ has parameters*

$$\boxed{n = q^m, \qquad k = \binom{m+d}{d}, \qquad \delta = 1 - \frac{d}{q}.}$$

Proof. For distinct polynomials P, Q of total degree at most d, apply Lemma 3 to $P - Q$. They disagree on at least a $1 - d/q$ fraction of points. This also proves injectivity of evaluation. To attain the bound, choose distinct $\alpha_1, \ldots, \alpha_d \in \mathbb{F}_q$ and take

$$P(X_1, \ldots, X_m) = \prod_{i=1}^{d} (X_1 - \alpha_i).$$

This polynomial is nonzero at exactly a $1 - d/q$ fraction of points. For $d = 0$, use the constant polynomial 1. □

The length can now exceed the alphabet size: $n = q^m$, while each symbol is still in $\mathbb{F}_q$. The rate is $\binom{m+d}{d}/q^m$. In particular, with q, d fixed and m growing, the rate tends to zero. We next consider the binary field and then allow the degree to be larger than the field size.

A Brief Look at the Binary Case

Over $\mathbb{F}_2$, every field element satisfies $x^2 = x$. Consequently, every function on $\mathbb{F}_2^m$ has a unique multilinear representation, and the code $\mathrm{RM}_2(m, d)$, for $0 \le d \le m$, consists of evaluations of

$$P(X) = \sum_{\substack{S \subseteq \{1,\ldots,m\} \\ |S| \le d}} c_S \prod_{i \in S} X_i.$$

Theorem 5 (Binary Reed–Muller parameters)**.** *For integers $m \ge 1$ and $0 \le d \le m$, the binary Reed–Muller code $\mathrm{RM}_2(m, d)$ has parameters*

$$\boxed{n = 2^m, \qquad k = \sum_{j=0}^{d} \binom{m}{j}, \qquad \delta = 2^{-d}.}$$

Proof. The block length is 2^m. There are $\sum_{j=0}^{d} \binom{m}{j}$ multilinear monomials of degree at most d, and uniqueness of multilinear representations makes evaluation injective. This gives the dimension.

Since the code is linear, its relative distance is the minimum fraction of nonzero coordinates in a nonzero codeword. Let $P \neq 0$ have degree $r \le d$. After relabeling variables, suppose $X_1 \cdots X_r$ is a monomial of P with nonzero coefficient. Fix $X_{r+1}, \ldots, X_m$ arbitrarily. The coefficient of $X_1 \cdots X_r$ remains 1: any other monomial containing all these variables would have degree greater than r. Thus the restricted polynomial is a nonzero multilinear polynomial in r variables. By uniqueness of multilinear representations, it has a nonzero evaluation at some choice of $X_1, \ldots, X_r$. For every assignment to the remaining variables, at least a 2^{-r} fraction of choices of the first r variables therefore give a nonzero evaluation. Hence

$$\Pr_{a \in \mathbb{F}_2^m}[P(a) \neq 0] \ge 2^{-r} \ge 2^{-d}.$$

Finally, $X_1 \cdots X_d$ is nonzero on exactly a 2^{-d} fraction of points, attaining the bound. For $d = 0$, take the constant polynomial 1. □

The Distance of General Reed–Muller Codes

The low-degree and binary distance bounds have a common generalization. We continue to use reduced polynomials, now allowing $d \geq q$.

Theorem 6 (General Reed–Muller distance)**.** *Let $m \geq 1$ and $0 \leq d < m(q-1)$. Write*

$$d = s(q-1) + t, \qquad 0 \leq t < q-1.$$

Then $\mathrm{RM}_q(m,d)$ has relative distance

$$\boxed{\delta = \frac{q-t}{q^{s+1}}.}$$

For $d \geq m(q-1)$, the code is all of $\mathbb{F}_q^{q^m}$ and has relative distance q^{-m}.

Duals of Reed–Muller Codes

For evaluation vectors, the standard inner product is

$$\langle \mathrm{Enc}(P), \mathrm{Enc}(Q) \rangle = \sum_{a \in \mathbb{F}_q^m} P(a)Q(a) \in \mathbb{F}_q.$$

The key fact is cancellation when we sum a low-degree monomial over all field elements.

Lemma 7 (Sums of powers)**.** *For $0 \leq j \leq q-1$,*

$$\sum_{a \in \mathbb{F}_q} a^j = \begin{cases} 0, & 0 \leq j < q-1, \\ -1, & j = q-1. \end{cases}$$

Here $a^0 = 1$, including when $a = 0$.

Proof. At $j = 0$, the sum is $q \cdot 1 = 0$ in $\mathbb{F}_q$. At $j = q-1$, every nonzero field element contributes 1, so the sum is $(q-1) \cdot 1 = -1$. For $0 < j < q-1$, let η generate $\mathbb{F}_q^*$. Multiplication by η permutes the field, giving

$$\sum_a a^j = \sum_a (\eta a)^j = \eta^j \sum_a a^j.$$

Since $\eta^j \neq 1$, the sum must be zero. □

Theorem 8 (Reed–Muller duality)**.** *For $0 \leq d < m(q-1)$,*

$$\boxed{\mathrm{RM}_q(m,d)^\perp = \mathrm{RM}_q\big(m, m(q-1) - d - 1\big).}$$

Proof. Write $M = m(q-1)$. First consider monomials X^e and X^f of total degrees at most d and $M - d - 1$, respectively. Then $\sum_i (e_i + f_i) \leq M - 1$, so some coordinate i satisfies $e_i + f_i < q - 1$. Factoring the sum and applying the lemma gives

$$\sum_{a \in \mathbb{F}_q^m} a_1^{e_1+f_1} \cdots a_m^{e_m+f_m} = \prod_{i=1}^{m} \left(\sum_{a_i \in \mathbb{F}_q} a_i^{e_i+f_i} \right) = 0.$$

By linearity, the two codes are orthogonal. Hence

$$\mathrm{RM}_q(m, M-d-1) \subseteq \mathrm{RM}_q(m,d)^{\perp}.$$

It remains to match dimensions. Complementing exponents,

$$(e_1, \ldots, e_m) \longmapsto (q-1-e_1, \ldots, q-1-e_m),$$

is a bijection from reduced monomials of total degree at least $d+1$ to those of total degree at most $M-d-1$. Therefore

$$K_q(m, M-d-1) = q^m - K_q(m,d) = \dim\big(\mathrm{RM}_q(m,d)^{\perp}\big).$$

Since $\mathrm{RM}_q(m, M-d-1)$ is contained in $\mathrm{RM}_q(m,d)^{\perp}$ and the two spaces have the same dimension, they are equal. □

Reminder: Familiar Binary Codes

Recall the following codes and duality relations from Problem Set 1.

Hadamard codes

For $d = 0$, $\mathrm{RM}_2(m,0)$ is the repetition code. For $d = 1$, the polynomials are affine linear functions

$$P_{b,u}(a) = b + u \cdot a, \qquad b \in \mathbb{F}_2, \quad u \in \mathbb{F}_2^m.$$

Thus $\mathrm{RM}_2(m,1)$ has parameters

$$n = 2^m, \qquad k = m+1, \qquad \delta = \frac{1}{2}.$$

These are the affine Hadamard codewords. Restricting to $b = 0$ gives the Hadamard code, of dimension m and relative distance $1/2$. Puncturing at $a = 0$ gives the punctured Hadamard code, with parameters

$$\boxed{n = 2^m - 1, \qquad k = m, \qquad \delta = \frac{2^{m-1}}{2^m - 1}.}$$

Extended Hamming and Hamming codes

For $m \geq 2$, duality gives

$$\mathrm{RM}_2(m,1)^{\perp} = \mathrm{RM}_2(m, m-2).$$

Its parameters are

$$\boxed{n = 2^m, \qquad k = 2^m - m - 1, \qquad \delta = 2^{2-m}.}$$

This is the *extended Hamming code*. Puncturing at $a = 0$ gives the *Hamming code*, with parameters

$$\boxed{n = 2^m - 1, \qquad k = 2^m - m - 1, \qquad \delta = \frac{3}{2^m - 1}.}$$

Its dual is the punctured Hadamard code.

Two Univariate Extensions

Both codes below have block length n and alphabet $\mathbb{F}_q^s$. The message polynomial has degree less than k. Relative distance is the fraction of coordinates at which the *whole tuples* differ.

Definition 9 (Order-s univariate multiplicity code)**.** Fix integers $s, n \geq 1$ and $1 \leq k \leq sn$, and distinct evaluation points $\alpha_1, \ldots, \alpha_n \in \mathbb{F}_q$. Assume $\mathrm{char}(\mathbb{F}_q) \geq s$. The order-$s$ univariate multiplicity code, UMULT, encodes $P \in \mathbb{F}_q[X]$ with $\deg P < k$ by

$$\mathrm{Enc}_{\mathrm{UMULT}}(P)_i = \big(P(\alpha_i), P'(\alpha_i), \ldots, P^{(s-1)}(\alpha_i)\big), \qquad 1 \leq i \leq n,$$

where $P^{(j)}$ is the usual jth formal derivative and $P^{(0)} = P$ [3].

Definition 10 (Order-s folded Reed–Solomon code)**.** Fix integers $s, n \geq 1$ and $1 \leq k \leq sn$, a nonzero $\gamma \in \mathbb{F}_q$ of multiplicative order greater than k, and evaluation points $\alpha_1, \ldots, \alpha_n \in \mathbb{F}_q$ satisfying, for every i, j and $0 \leq t < s$,

$$\alpha_i \gamma^t = \alpha_j \quad \Longleftrightarrow \quad i = j \text{ and } t = 0.$$

For example, one may take γ to be a primitive element of $\mathbb{F}_q$ with $q - 1 > k$, while retaining the separation condition above. The order-s folded Reed–Solomon code, FRS, encodes $P \in \mathbb{F}_q[X]$ with $\deg P < k$ by

$$\mathrm{Enc}_{\mathrm{FRS}}(P)_i = \big(P(\alpha_i), P(\alpha_i\gamma), \ldots, P(\alpha_i\gamma^{s-1})\big), \qquad 1 \leq i \leq n.$$

The separation condition makes all sn evaluation points distinct [2, Section 17.1].

Theorem 11 (Rate and relative distance of UMULT and FRS)**.** *Under their respective hypotheses, both order-s codes have q^k codewords, block length n, alphabet $\mathbb{F}_q^s$, and*

$$\boxed{R = \frac{k}{sn}, \qquad \delta = 1 - \frac{1}{n}\left\lfloor \frac{k-1}{s} \right\rfloor.}$$

References and Further Reading

[1] Prahladh Harsha. Coding Theory, Lecture 8. Tata Institute of Fundamental Research, September 21, 2022. Reed–Muller portion, pages 3–8.

[2] Venkatesan Guruswami, Atri Rudra, and Madhu Sudan. Essential Coding Theory. Draft of August 26, 2025. Sections 9.1–9.4 for Reed–Muller codes and their distance, and Section 17.1 for folded Reed–Solomon codes.

[3] Swastik Kopparty. Some remarks on multiplicity codes. arXiv:1505.07547, 2015.